



ECE3403 - Data Analytics & Optimization

Term Project

Lecturer

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1. PROJECT DESCRIPTION

Form a team of four students to work on a selected dataset from an active machine learning competition. Each team will:

- Explore and preprocess the dataset.
- Identify and solve a key problem using supervised or unsupervised learning.

Each problem will be assigned to 3–4 teams. Assignments will be randomized across the available competitions (*will be announced later*)

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2. PROJECT DELIVERABLES

Each team must complete the following:

1. **Data Processing**

- Handle missing values, outliers, and class imbalance.
- Apply appropriate transformations.

2. **Handle one or two of these problems in the dataset**

3. **Data Exploration**

- Include thorough exploratory analysis and visualizations.

4. **Model Implementation**

- Use a suitable supervised or unsupervised machine learning model.

5. **Report**

- Compare results before and after preprocessing.
- Final Report must include:
 - Introduction
 - Motivation & Objectives
 - Dataset Description
 - Evaluation Metrics
 - Results

6. **Bonus**

- Bonus marks are available for either or both of the following:
 - Correctly implementing and evaluating additional or advanced techniques on the dataset (Must show before and after).
 - Achieving the highest model metrics among all teams working on the same dataset

3. SUBMISSION REQUIREMENTS

Submit the following by the deadline:

1. **Code (fully functional, with clear explanations for each step)**

2. **Report (structured and complete)**

3. **Submit predictions to the competition platform**

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4. EVALUATION CRITERIA

1. **Implementation (Code: 6 marks)**
 - a. Fulfilling requirements **(3.5 Marks)**.
 - b. Individual Discussion **(2.5 Marks)**.
2. **Documentation (Report: 4 marks)**
 - a. Fulfilling requirements **(2 Marks)**.
 - b. Plagiarism/AI Score **(2 marks)**.

Plagiarism Score	Marks
0% → 20%	2
20% → 40%	1
More than 40%	0

5. DISCUSSION GUIDELINES

- All team members must attend the project discussion.
- No late submissions accepted.
- Plagiarism results in zero marks (0/10) for the entire team.

6. DEADLINS

- **Submission (Code & Report): Sunday, 31 May 2026**
(Submit on both **Google Classroom** and **Competition platform**)
- **Project Discussions:** Monday & Tuesday, **1–2 May 2026 (Week 15)**
(Discussion slots will be announced later)